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NPS/001/004 Technical Specification for 11kV, 20kV and 33kV Pole Mounted Expulsion Fuses, Auto Sectionalising Links and Solid Links

1. Purpose

The purpose of this document is to detail the technical requirements for 11kV, 20kV and 33kV drop-out expulsion fuse mounts, fuse carriers, automatic sectionalising links and solid links for use on the CE Electric UK distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/011/011	1.0	Specification for 11kV Expulsion Fuse Switches
DSS/011/001	1.0	Specification for 11kV Pole Mounted Auto sectionalising Switch Disconnecter
NE FD1	2.0	Specification for 11,20 & 33kV Expulsion Fuse Units
NPS/001/004	1.0	Specification for 11, 20 and 33kV Pole Mounted Expulsion Fuses, Auto Sectionalising Links and Solid Links

2. Scope

This specification details the range, design, constructional and technical requirements for 11kV, 20kV and 33kV pole mounted expulsion fuses, automatic sectionalising links and solid links.

3. Technical Specification

3.1 General Requirements

Pole-mounted fuse gear is used widely to protect distribution transformers and HV overhead line circuits and to provide a means of isolation for operational purposes. Automatic Sectionalising Links (ASL) and Solid Links can be used in place of fuse carriers in drop-out fuse mounts. Consequently these links and mount combinations are required to have a rated short-time withstand current value adequate for circuits controlled by circuit breakers equipped with inverse definite minimum time lag (IDMTL) protection.

All combinations are used with dependent manual operation, within the limits specified to disconnect live circuits and to energise circuits believed to be healthy.

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Equipment should comply generally with ENA TS 41-36: 2004 and with the relevant standards referred to therein. CE Electric UK have a preference for a single unit to satisfy both 11kV and 20kV requirements utilising composite silicon insulators.

The equipment is required to provide: -

- a) automatic disconnection and isolation of supply in the event of a fault occurring beyond the point of installation.
- b) load-breaking disconnection - Manual
- c) fault-making reconnection - Manual

Appendix 2 details the range of products required for this specification.

3.2 Fuse Mounts

3.2.1 Mounting Arrangement

The equipment shall be designed for manual operation using insulating operating poles not less than 4.8m long inclined at an approx. angle of 15° to the vertical. The operations required are insertion, removal, opening and closing. The maximum force applied to the link during closing is 150N.

The equipment shall comprise a vertically mounted, single pole, with the option to install removable expulsion type fuse carriers, ASL's or solid links.

The equipment shall be suitable for mounting on the 102mm web of a channel or equivalent surface having a 14mm slotted fixing hole located with centre line between 45mm and 60mm from the upper edge. Single hole fixing arrangements shall have provision to prevent fuse mount swivelling.

The lower fixed contact assembly shall be so designed as to permit the fuse-carrier, ASL or solid link to be engaged as effortlessly as possible, using an appropriate insulated operating pole in severe weather conditions.

3.2.2 Fault Mode Operation

The equipment shall be so designed and constructed such that rupturing of the fuse element, or operation of the ASL will cause the upper contact of the unit to disengage from the upper fixed contact of the fuse-mount. This shall allow the fuse-carrier or ASL to pivot under the influence of gravity to about a horizontal axis on the lower fixed contact of the fuse-mount, and come to rest hanging vertically, or near vertically, from the lower fixed contact of the fuse-mount.

3.2.3 Conductor Terminations

The upper and lower fixed contacts of the fuse-mount shall be equipped with suitable terminals to enable stranded overhead line copper conductors in the range 32mm² to 125mm² to be securely connected to them. The terminals shall be supplied complete with studs or bolts and nuts with plain

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and lock washers suitable for the termination of either a lug with a single hole palm, 14mm hole or alternatively to clamp stranded copper conductor.

3.2.4 Contact Fittings

The upper fixed contact assembly shall be so designed, and if necessary equipped with appropriate and suitable guides, as to ensure as far as is reasonably practicable that the fuse-carrier, ASL or solid link will correctly engage during all closing operations under all service conditions.

The fixed contacts of the fuse-mount shall be such that a positive, high pressure, non-deteriorating electrical contact with the fuse-carrier, ASL or solid link is maintained under all service conditions.

Contacts shall provide a wiping/cleaning action on closing.

3.2.5 Interchange ability

The upper and lower portions of the fuse shall have a shorting bar attached that will allow the fitting of a 'shorting device'. This will allow the interchanging of fuses, solids and sectionalisers thus reducing disruption to customer's supplies.

3.2.6 Additional Requirements

The mounting units shall have a maximum weight of 12.5kg.

The mounting unit shall permit the fuse-link, ASL or solid links to be inserted and secured without the need for any additional special tooling.

The mounting unit shall be designed and constructed to prevent the accumulation of water within it under all conditions of service.

The unit shall be capable of being operated using the standard insulated operating pole and head currently in use within CE Electric UK.

The fuse-carrier shall be of the expulsion type capable of accepting fuse-links having ratings of 6 to 100 amps.

3.3 Ratings

The complete fuse assembly shall have the following minimum ratings that shall be verified by appropriate type tests carried out in accordance with the requirements of IEC 60282 Part 2.

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Voltage (kV)	Equipment Type	Rated Current (A)	Rated Short-time Withstand Current (kA)	Rated Peak Withstand Current (kA)	Rated Duration of Short Circuit (seconds)	Breaking capacity (kA)
12	Expulsion Fuse	100, 200 (with Fuse)	N/a	N/a	N/a	8.0
	Solid	200	4	10	3	N/a
	ASL	20, 25, 30, 36, 40, 56, 80, 112 *	4	10	3	N/a
24	Expulsion Fuse	100, 200 (with Fuse)	N/a	N/a	N/a	8.0
	Solid	200	4	10	3	N/a
24	ASL	20, 25, 30, 36, 40, 56, 80, 112 *	4	10	3	N/a
36	Expulsion Fuse	100, 200 (with Fuse)	N/a	N/a	N/a	4
	Solid	200	4	10	3	N/a

* Values for ASL's are minimum actuating current values.

3.4 Automatic Sectionalising Links

3.4.1 General Requirements

Automatic sectionalising links shall be designed to be physically interchangeable with the fuse carriers and solid links of drop-out expulsion fuses. They are required for both ANSI and J & P type back units. The ASL's are designed to recognise a pre-determined sequence or fault currents and then, during a period when the upstream protective device is open, to disengage and drop-out to the isolating position.

The ASL's shall be interchangeable with the fuse carrier and have a ratings available of between 20 and 112 amps. The actuator in this device is required to trip in the 'dead time' after two or three passages of fault current. Actuators shall preferably be of the electronic re-settable type with a selection of ratings available for each unit to reduce the number items stocked.

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3.4.2 Minimum Actuating Current

The table below details the minimum actuating current required. These ratings shall be available from the range of two ASL's.

Minimum Actuating Current	Unit
20A	Type 1 ASL
25A	Type 1 ASL
30A	Type 1 ASL
36A	Type 1 ASL
40A	Type 2 ASL
56A	Type 2 ASL
80A	Type 2 ASL
112A	Type 2 ASL

3.4.3 Reclaim Time

ASL devices shall have a reclaim time of 25 seconds.

3.4.4 Number of Counts

ASL devices are designed to be set to operate in at least one less count than the upstream pole mounted auto recloser. Units are required to be capable of either two or three shot operation.

3.5 Insulation

The preferred rated voltage and insulation levels for this equipment shall be as detailed in ENA TS 41-36 Table 1.1b. (CE Electric UK will accept a reduced (common value) lightening impulse withstand voltage of 125kV where 150kV is not offered as a standard option).

3.6 Electrical Clearances

A test voltage shall be applied across the terminals of the fuse mount with the carrier removed in accordance with BS 2692-part 2 clause 22 to confirm that the fuse mount is capable of providing isolation properties in line with the electrical values stated in clause 3.5 above. Additionally fuse mounts shall be designed to comply with the "basic electrical clearance" (Phase to Earth) values detailed in BS7354 1990 "Design of HV Open Terminal Stations" Table 4 associated with the nominated BIL of the equipment.

Gaps between live and earthed metalwork of complete equipment's shall be adequate to avoid short circuits by the presence of the operating rod.

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3.7 Testing

3.7.1 Type Tests - General

Complete assemblies shall comply with BS2692 except that the requirements for rated breaking capacity shall not apply to automatic sectionalising links and solid links.

3.7.2 Additional Clauses for Sectionalising Links

Type tests for automatic sectionalising links shall be as detailed ENA TS 41-36 Clause 8.6.3. Additional tests shall provide evidence of tests carried out in corrosive atmosphere, salt-laden mists and other environmental hazards not specifically above.

3.7.3 Routine Tests

Routine tests shall be carried out in accordance with BS 2692 Part 2 - Clause 2.9.

4. References

4.1 External Documentation

Reference	Title
ENA TS 41-36: 2004	"Specification for Distribution Switchgear for Service up to 36 kV (Cable and Overhead Line Conductor Connected)".
BS 7354 : 1990	Code of practice for design of high-voltage open-terminal stations
BS 2692 Part 2: 1956 / IEC60282 Part 2: 1997	"Fuses for voltages exceeding 1000V a.c. – Expulsion Fuses"

4.2 Internal documentation

Reference	Title

5. Definitions

Term	Definition

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	26/11/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	26/11/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead System Standards Manager	G Hammel	26/11/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	14/12/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	27/11/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	27/11/07

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Appendix 1 – Self Certification Conformance Declaration – Common Clauses

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 1 and the clauses of the aforementioned Standards relevant to common specifications for high-voltage switchgear and control gear standards. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/appropriate to the product

Cs1 = The test conforms fully with the requirements of this clause

Cs2 = The test conforms partially with the requirements of this clause

Cs3 = The test does not conform to the requirements of this clause

Cs4 = Test not performed, but alternative evidence/ technical case offered

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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IEC 62271-102, B.S. 2692:Part 2				ENATS 41-36 Part 8			Remarks
Clause / Sub-clause IEC 62271-102 BS 2692 : Part2		Requirement	Confor - mance code	ENATS 41- 36 - Part 1 Clause / Sub- clause	Requirement	Confor - mance code	
1	Section 1	General		8.1	General		
2	Section 4	Normal and special service conditions		8.2	Normal and special service conditions		
3	Section 1	Definitions		8.3	Definitions		
4	Section 2	Ratings		8.4	Ratings		

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IEC 62271-102, B.S. 2692:Part 2				ENATS 41-36 Part 8			Remarks
Clause / Sub-clause IEC 62271-102 BS 2692 : Part2		Requirement	Confor - mance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Confor - mance code	
4.1	Section 2 cl4	Rated voltage		8.4.1	Rated voltage		
4.2		Rated insulation level		8.4.2	Rated insulation level		
4.3		Rated frequency		8.4.3	Rated frequency		
4.4	Section 2 cl5	Rated current		8.4.4	Rated current / minimum actuating current		
4.5		Rated short-time withstand current		8.4.5	Rated short-time withstand current		
4.6		Rated peak withstand current		8.4.6	Rated peak withstand current		
4.7		Rated duration of short-circuit		8.4.7	Rated duration of short-circuit		
	Section 2 cl6			8.4.8	Breaking capacity		
				8.4.9	Reclaim time		
	Section 3	Marking					
	Section 5	Operation in service					
	Section 6	Construction		8.5	Design and construction		
				8.5	General		
				8.5	Manual operation min 4.8m at 15°		
				8.5	Max force 150N		
				8.5	Clearance for operating pole head		
				8.5	Pivot interlock		
				8.5	Operating eye/hook		
				8.5	Fuse link length tolerance		

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	Section 6	Construction		8.5	Robust/corrosion free links		
				8.5	Link adjustment instructions		
				8.5	Ease of replacement of actuators		

IEC 62271-102, B.S. 2692:Part 2				ENATS 41-36 Part 8			
Clause / Sub-clause IEC 62271-102 BS 2692 : Part2		Requirement	Confor - mance code	ENATS 41- 36 - Part 1 Clause / Sub- clause	Requirement	Confor - mance code	Remarks
	Section 6	Construction		8.5.2	Mounting arrangement		
				8.5.3	Terminals		
				8.5.4	Drop-out automatic sectionalising links		
				8.5.5	Marking of voltage rating		
				8.5.6	Labelling -information (a) to (g)		
	Section 7	Type tests		8.6	Type testing		
	Section 7 cl 20	General		8.6.1	General - Table 8.2		
	Section 7 cl 21	Test for temperature rise					
	Section 7 cl 22	High-voltage test					
	Section 7 cl 23	Impulse-voltage test					
	Section 7 cl 26	Weatherproofing test					
	Section 7 cl 27	Test for breaking-capacity					

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IEC 62271-102, B.S. 2692:Part 2			ENATS 41-36 Part 8			Remarks
Clause / Sub-clause IEC 62271-102 BS 2692 : Part2	Requirement	Confor - mance code	ENATS 41- 36 - Part 1 Clause / Sub- clause	Requirement	Confor - mance code	
	Section 7 cl 28	Criteria of failure in tests for breaking capacity				
			8.6.2	Tensile test		
			8.6.3	Type test for automatic sectionalising links		
			8.6.3.1	Operation tests (a) to (g)		
			8.6.3.2	Short-time withstand current test		
			8.6.3.3	Special tests		
			8.6.3.4	Performance characteristics		
	Section 8		8.7	Routine tests		

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IEC 62271-102, B.S. 2692:Part 2				ENATS 41-36 Part 8			
Clause / Sub-clause IEC 62271-102		Requirement	Confor - mance code	ENATS 41-36 - Part 1 Clause / Sub-clause	Requirement	Confor - mance code	Remarks
9				8.8	Information to be given with enquiries, tenders and orders - schedule 8.1		
10				8.9	Rules for transport, storage, erection, operation and maintenance		
11				8.10	Safety		

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Appendix 1 Cont. - SELF CERTIFICATION CONFORMANCE DECLARATION

Type tests for overhead conductor connected expulsion fuses, solid links and automatic sectionalising links.

Switchgear covered by ENATS 41-36 shall comply with the latest issues of the relevant International and British Standards. ENATS 41-36 is intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses in ENATS 41-36 - Part 1 and the clauses of the aforementioned Standards relevant to common specifications for high-voltage switchgear and control gear standards. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

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- Prefix each remark with the relevant 'IEC' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
	Test Requirement	Specification and standards	Rated value	Test req'd Y or N	Conformance *	Test value	Date of test	Test station Report / Cert No	Witness I, M or ENA **	Remarks
1	High-voltage test	B.S. 2692 : Part 2, Clause 22 Table 1.1b of ENATS 41-36								
2	Impulse-voltage test	B.S. 2692 : Part 2, Clause 23 Table 1.1b of. ENATS 41-36								
3	Temperature Rise	B.S. 2692 : Part 2, Clause 21								
4	Weatherproof test for powder-filled cartridge fuses (where applicable)	B.S. 2692 : Part 2, Clause 26								
5	Test for breaking capacity ***	B.S. 2692 : Part 2, Clause 27								
6	Short-time withstand current and peak	IEC 62271-102. Sub-clause 6.6.								
7	Verification of protection	IEC 62271-102. Sub-clause 6.7, Sub-clause 1.5.13 of ENATS 41-36.								
8	Tensile test on expulsion fuse-link	Test method described in sub-clause 8.6 2 of ENATS 41-36								
9	Operational test for automatic sectionalising links	Test method described in sub-clause 8.6 3 of ENATS 41-36								

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Appendix 2 – Schedule of Products

Commodity Code	Description
245944	11kV/20Kv Back Portion
tba	33kV Back Portion
246947	11/20kV Fuse Tube
tba	33kV Fuse Tube
245164	11kV/20kV Solid Link
247541	11/20kV Automatic Sectionalising Link for ANSI Standard Unit. Type 1. Ratings available – 20, 25, 30, and 36A
247515	11/20kV Automatic Sectionalising Link for ANSI Standard Unit. Type 2. Ratings available – 40, 56, 80 and 112A
247516	Automatic Sectionalising Link for 11kV J & P Unit. Type 1. Ratings available – 20, 25, 30, and 36A
247517	Automatic Sectionalising Link for 11kV J & P Unit. Type 1. Ratings available – 40, 56, 80 and 112A